PUNJAB ENGINEERING COLLEGE (DEEMED TO BE UNIVERSITY), CHANDIGARH

MINOR PROJECT PROGRESS REPORT

OPTIMIZATION-BASED RESOURCE ALLOCATION FOR PUBLIC BUS FLEET SCHEDULING UNDER DEMAND VARIABILITY

(A case study of the Chandigarh Transport Undertaking)

Submitted in partial fulfilment of the requirements for the award of the degree of

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Under the guidance of

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Declaration

We hereby declare that the work presented in this progress report, entitled "Optimization-Based Resource Allocation for Public Bus Fleet Scheduling under Demand Variability", is our own work carried out under the guidance of Dr. Sumati Mahajan, Department of Mathematics, Punjab Engineering College, Chandigarh. The work has not been submitted elsewhere for the award of any degree. Sources of information have been acknowledged in the references.

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Abstract

Urban bus operators must allocate a fixed fleet across many routes whose demand varies by place, time of day and day to day. This project formulates that allocation as a multi-objective optimization problem and applies it to the Chandigarh Transport Undertaking (CTU), using its published timetable: 68 routes, 389 allocated buses and 1,662 km of route length. The data were audited against the timetable PDF, and cycle times were calibrated so that the published frequencies are consistent with the allocated fleet.

The allocation is formulated as a mixed-integer linear program (651 variables, 576 constraints) and solved exactly: the waiting-time objective is linearised with a discrete headway menu, operating effort is measured in vehicle-kilometres, hourly capacity is bus capacity times frequency, and demand uncertainty uses the upper α-cut of a triangular fuzzy number. Goal programming with explicit priorities is used to balance waiting, vehicle-km, overcrowding and headway equity, and the trade-offs among the three objectives are traced on an exact AUGMECON Pareto front computed from the same MILP.

No ridership data are public, so demand is synthetic and illustrative. Under this demand, with the maximum frequency relaxed and $\alpha = 0.5$, the balanced goal-programming schedule reduces expected waiting by 4.1%, vehicle-km by 3.1% and overflow passengers by 26.4% relative to the published schedule, using the 357 buses allowed by the model's spare rule. The results are not measured CTU outcomes: they demonstrate the method, and the report states where they could be artefacts of the synthetic demand and of the calibration. Metaheuristic search (NSGA-II, PSO and QPSO) and Monte Carlo simulation are left as future work.

Keywords: Bus Fleet Allocation, Frequency Setting, Mixed-Integer Linear Programming, Goal Programming, Fuzzy Demand, α-Cut, Pareto Front, AUGMECON.

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Chapter 1 Introduction

1.1 General

Fixed-route bus services are the most accessible and affordable form of public transport for most Indian city commuters. A transport undertaking runs a finite fleet over a network of routes whose demand changes across the city, across the day, and from one day to the next. When the planned supply of departures does not follow this demand, the network fails in two ways at once: passengers on busy corridors face long waits and crowded or full buses, while buses on quiet corridors run with spare capacity that still costs fuel, maintenance and driver hours.

From an applied-mathematics point of view, this is a constrained multi-objective resource-allocation problem: a limited pool of buses must be divided among routes and time periods to satisfy objectives that conflict. More frequent service reduces waiting and crowding but increases vehicle-kilometres; fewer runs save cost but hurt passengers. A planning method must therefore expose the trade-offs, not just one "optimal" answer [1, 4].

1.2 Background of the Project

The case study is the Chandigarh Transport Undertaking (CTU), which operates local and tri-city services across Chandigarh, Mohali and Panchkula. The input is CTU's published timetable for Depots II, III and IV (as on 25.07.2026) [25]: for each of the 68 routes it gives the operating hours, the published headway range, the route length and the number of buses allocated (389 in total). A stop-coverage list adds 649 named stops.

The project began from a synopsis that proposed a multi-objective model with three objectives (waiting time, operating effort and overcrowding), fleet and frequency constraints, fuzzy demand and three solution approaches (MILP, NSGA-II and fuzzy linear programming). During execution, four mathematical errors in that formulation were found and corrected (Section 3.3). The solution methods were then restricted to the linear (exact) ones, with heuristic search left as future work. The most important practical constraint is that CTU publishes no ridership data, so all passenger demand in this report is synthetic and illustrative; the methods are built so that real ridership can replace it without changing the model.

1.3 Objectives

1. Build a validated baseline of the CTU network from the published timetable (routes, headways, buses, stops) and calibrate cycle times against the allocated fleet.
2. Formulate a linear multi-objective mixed-integer model for choosing headways and allocating buses by route and time period.
3. Represent uncertain demand with triangular fuzzy numbers and an α-cut capacity rule.
4. Balance passenger waiting, operating effort, overcrowding and headway equity using goal programming with explicit priorities, and trace the trade-offs on an exact Pareto front

(AUGMECON) computed from the same MILP.

5. State honestly where the results hold and where they break (demand assumptions, calibration and modelling simplifications), and set out the extensions (metaheuristic search, Monte Carlo simulation and real ridership data) as future work.

Chapter 2 Literature Review

2.1 Review of Research Papers

Transit planning and frequency setting

Bus planning is classically decomposed into network design, frequency setting, timetabling, and vehicle and crew scheduling [1, 2]; Ibarra-Rojas et al. [4] review the planning, operation and control problems in this chain. Furth and Wilson [3] formulated the allocation of a fixed fleet across routes and time periods subject to fleet-size and loading constraints, which is the structure of the present model. Martínez et al. [5] formulate frequency setting as a mixed-integer program over a discrete set of frequencies, and Verbas and Mahmassani [6] allocate frequencies over routes and time periods while exposing the trade-offs explicitly.

Multi-objective transit optimization

When routes and frequencies are designed together, the search space is very large and metaheuristics dominate: Arbex and da Cunha [17] use an alternating-objective genetic algorithm, Jha et al. [18] a multi-objective particle swarm compared with NSGA-II, and Ahern et al. [19] warn that fleet-minimising designs that ignore passenger response can produce overcrowding. NSGA-II [7] is the standard elitist multi-objective genetic algorithm. For exact fronts, the augmented ε-constraint method AUGMECON [8] avoids weakly efficient solutions, and AUGMECON2 [9] is designed for integer programs.

Goal programming

Goal programming originates with Charnes and Cooper [10]. Each objective becomes a goal with a target, deviation variables measure under- and over-achievement, and only the unwanted deviations are minimised, either in strict priority order (preemptive) or as a weighted sum of normalised deviations [11]. The method makes priorities explicit, but its answer depends on the targets and on the normalisation.

Demand uncertainty: stochastic and fuzzy approaches

Stochastic programming handles uncertain demand when distributions can be estimated, for example in transit network design [15] and frequency setting [16]. When data are too scarce to fit distributions, fuzzy sets [12] represent demand by a few expert values; Zimmermann [13] solved fuzzy multi-objective linear programs, and possibilistic linear programming [14] converts fuzzy coefficients into crisp constraints through α-level cuts. The present project uses this α-cut route.

Service standards and the Indian context

Transport for London plans bus service levels against published guidelines [21], and Singapore's bus service reliability framework rewards operators according to Excess Wait Time [20]; this report uses EWT as a reliability measure in its future simulation work. GPS studies of Indian city buses report average speeds of 17.8 km/h in the peak and 21.5 km/h off-peak in Chennai [22] and strong spatial and temporal variation in Ahmedabad [23]. A review of crowding valuation [24] shows that crowding costs are context-specific, so this project does not convert crowding into money.

Table 1: Summary of the reviewed literature and its use in this project

Ref.	Topic	Use in this project
[1]	Transit planning chain (textbook)	Frames frequency setting between network design and timetabling
[2]	Bus network design	Network design stage that precedes frequency setting
[3]	Fleet allocation across routes and periods	Same structure as our fleet and loading constraints
[4]	Review of bus planning, operation and control	Places this model within the planning chain
[5]	MILP frequency setting with discrete frequencies	Supports the headway-menu linearisation
[6]	Frequency allocation over routes and periods; trade-offs	Period-wise allocation and trade-off analysis
[7]	NSGA-II	Planned heuristic Pareto front, to be benchmarked against the exact one (future work)
[8, 9]	AUGMECON / AUGMECON2	Exact, strictly efficient ε-constraint front
[10, 11]	Goal programming (origin; practice)	Targets, deviations, preemptive and weighted forms
[12, 13, 14]	Fuzzy sets; fuzzy and possibilistic LP	Triangular fuzzy demand and α-cuts
[15, 16]	Transit design and frequency setting under demand uncertainty	Stochastic alternatives to the fuzzy approach
[17, 18, 19]	Multi-objective route design and frequency setting (GA, PSO)	User-operator Pareto fronts; overcrowding risk
[20, 21]	Bus service planning; Excess Wait Time in Singapore	Service-quality metric (EWT), for future simulation work
[22, 23]	Measured bus speeds in Indian cities	Speed range 18–26 km/h used in calibration
[24]	Crowding valuation review	Why crowding costs are not transferred between cities
[25]	CTU published timetable	Source of routes, headways, route lengths and buses

2.2 Research Gaps

- **Goal programming in bus frequency and fleet allocation.** A Crossref search of peer-reviewed work from 2014 to 2026 found no study applying goal programming to bus frequency setting or fleet allocation; the goal-programming transportation papers found address the classical (shipping) transportation problem. This is a statement about our search, not a claim that no such study exists.

- **No public ridership.** CTU publishes no ridership data, so the method must work with fuzzy demand and must be ready to take real data later.
- **Exact fronts at realistic size.** Many frequency-setting studies use only metaheuristics; for a fixed-route network of this size, the exact MILP can be solved directly and used to trace the whole Pareto front, which also provides the benchmark against which heuristics can be judged in future work.

Chapter 3 Methodology

3.1 Block Diagram

Figure 1 shows the pipeline. Stages 1 to 6 are the linear-model pipeline that was implemented; stage 7 (dashed) lists the next steps for extending the work. A single command (`python run_all.py`) runs the stages in order and regenerates the tables and figures of this report.

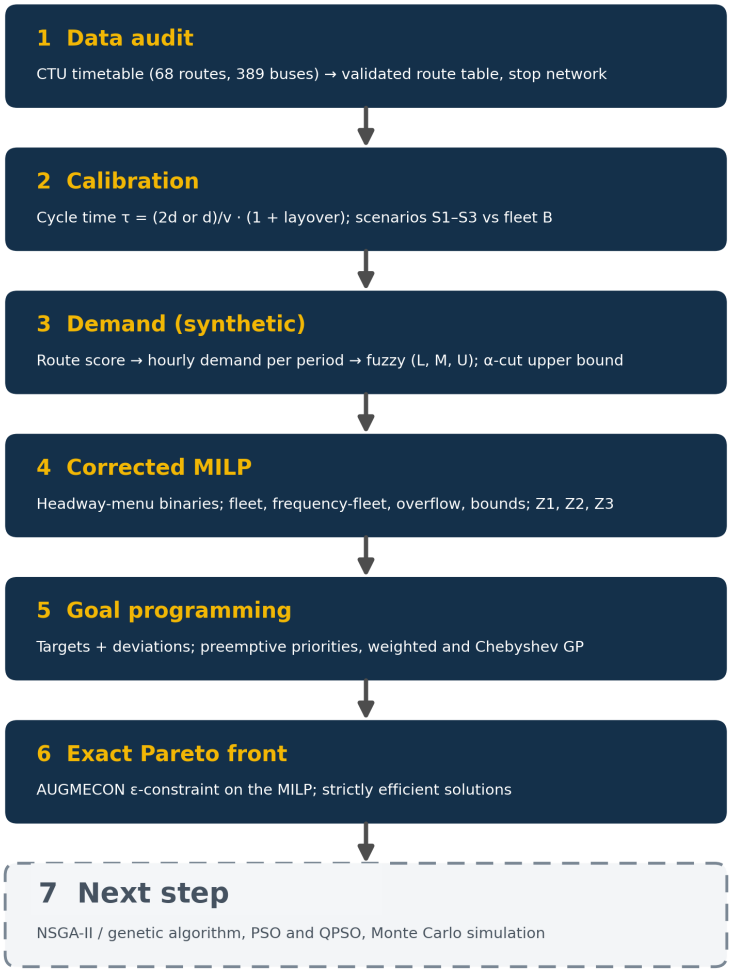


Figure 1: Methodology block diagram (stages 1–6 implemented; stage 7 shows the next step)

3.2 Data Preparation and Calibration

The timetable workbook was parsed into a clean route table: the TOTAL and summary rows were dropped, route numbers were kept as strings (for example "25/25A"), and frequencies were computed exactly as 60 divided by the headway. All 68 routes match the timetable PDF; 17 rounded cells in the original workbook were replaced by exact values. 8 routes are loops (origin equals destination) and 3 are overnight services.

The cycle time of route r (time for one bus to return to its starting point) is

$$\tau_r = \frac{L_r}{v}(1+\ell), \qquad L_r = 2d_r \text{ (non-loop) or } d_r \text{ (loop)} \tag{3.1}$$

where v is the average operating speed and ℓ the layover fraction. By Little's law, a route running f departures per hour in each direction needs $f \cdot \tau$ buses. Three readings of the published length were compared against the fleet (Section 5.1). Only the one-way reading makes the published timetable consistent with the allocation, and the working setting is $v = 22$ km/h with 10% layover. This is a consistency argument, not a proof; CTU must confirm the length convention.

3.3 Mathematical Model

Sets, variables and parameters. Routes $r \in R$ (68), periods $t \in T$ (AM 07–10, Midday 10–16 and PM 16–19), and for each active route-period a headway menu K_{rt} built from the values 10, 12, 15, 20, 25, 30, 40, 45, 60 and 90 minutes plus the route's published headways. Decision variables: binary y_{rtk} (route r uses headway k in period t), integer buses x_{rt}, and overflow $o_{rt} \geq 0$. Parameters: modal demand M_{rt} and its α-cut D^{α}_{rt} (passengers per hour), service hours H_{rt}, cycle distance d^{cyc}_r, bus capacity $C = 60$, and $B_{\text{eff}} = 357$ buses (spare ratio 8%).

The three objectives, all minimised, are:

$$Z_1 = \sum_{r,t} M_{rt} \sum_k \frac{h_k}{2} y_{rtk} \tag{3.2}$$

$$Z_2 = \sum_{r,t} d^{\text{cyc}}_r H_{rt} f_{rt} \tag{3.3}$$

$$Z_3 = \sum_{r,t} o_{rt} H_{rt} \tag{3.4}$$

Z_1 is the expected waiting rate (passenger-minutes per hour, half the headway per passenger), Z_2 the operating effort (vehicle-km per day) and Z_3 the number of passengers above capacity per day. All three are linear in the decision variables. The objectives are subject to:

$$\sum_k y_{rtk} = 1, \qquad f_{rt} = \sum_k \frac{60}{h_k} y_{rtk} \tag{3.5}$$

$$\sum_r x_{rt} \leq B_{\text{eff}} \qquad \forall t \tag{3.6}$$

$$\tau_r f_{rt} \leq x_{rt}, \qquad x_{rt} \in \mathbb{Z}_{\geq 0} \tag{3.7}$$

$$o_{rt} \geq D^{\alpha}_{rt} - C f_{rt}, \qquad o_{rt} \geq 0 \tag{3.8}$$

$$f^{\min}_r \leq f_{rt} \leq f^{\max}_r \tag{3.9}$$

$$|x_{rt} - x^0_r| \leq \max\bigl(1,\ \min(3,\ 0.3\,x^0_r)\bigr) \tag{3.10}$$

Equation (3.5) selects exactly one headway for every active route-period and defines its frequency; (3.6) is the fleet limit in every period, with $B_{\text{eff}} = \lfloor (1-s)B \rfloor$; (3.7) gives the buses needed to run the chosen frequency; (3.8) measures overflow above the hourly capacity $C \cdot f$; (3.9) keeps frequencies within the published bounds through the menu K_{rt}; and (3.10) limits changes from the published allocation. The 60/90-minute headway cap is treated as a soft goal.

The model has 191 active route-periods, 651 variables and 576 constraints, and solves to proven optimality in well under a second. Its correctness was checked against brute-force enumeration on a small test network. Table 2 lists the corrections relative to the synopsis.

Table 2: Corrections made to the synopsis formulation

Synopsis	Problem	Correction
$Z_1 = \sum D_{rt} \cdot 30/f_{rt}$	Nonlinear in f (division by a decision variable)	Headway menu with binaries y_{rtk}; $Z_1 = \sum M_{rt} \sum_k (h_k/2)\, y_{rtk}$ is linear
$Z_2 = \sum d_r \cdot x_{rt}$	Counts buses, not distance run	Vehicle-km $Z_2 = \sum d_r^{\text{cyc}} H_{rt} f_{rt}$
$o_{rt} \geq D_{rt} - C \cdot x_{rt}$	Capacity per hour depends on frequency, not on buses	$o_{rt} \geq D_{rt}^{\alpha} - C \cdot f_{rt}$
$Cx \geq$ $D^L + (1-\alpha)(D^M - D^L)$	Lower α-cut: plans for low demand; probability statement used for a fuzzy number	Upper α-cut $D^{\alpha} = U - \alpha(U - M)$; possibility ($\alpha$-level) interpretation
f_{rt} continuous	Published headways come in discrete steps	Discrete headway menu K_{rt} (route-specific)

Two published constraints made the brief infeasible as first stated, and were adjusted with reasons recorded in the project's decision log: with 10% spares, the published minimum frequencies need at least 352 buses but only 350 would be available, so the spare ratio is 8%; and the practicality rule $|x - x^0| \leq \min(3, 0.3x^0)$ froze every small route, so a minimum change of one bus is allowed.

3.4 Fuzzy Demand and the α-Cut

Demand on each route-period is a triangular fuzzy number (L, M, U) with $L = 0.80M$ and $U = 1.35M$, right-skewed because demand surges are larger than dips. For a confidence level $\alpha \in [0, 1]$, the upper end of the α-cut is

$$D_{rt}^{\alpha} = U_{rt} - \alpha\,(U_{rt} - M_{rt}) \tag{3.11}$$

Capacity is planned against this value: $\alpha = 1$ plans for the most plausible demand M, and $\alpha = 0$ for the most conservative value U. The waiting objective uses M, because it measures the expected passenger experience; the capacity constraint uses D^{α}, because it is a safety requirement.

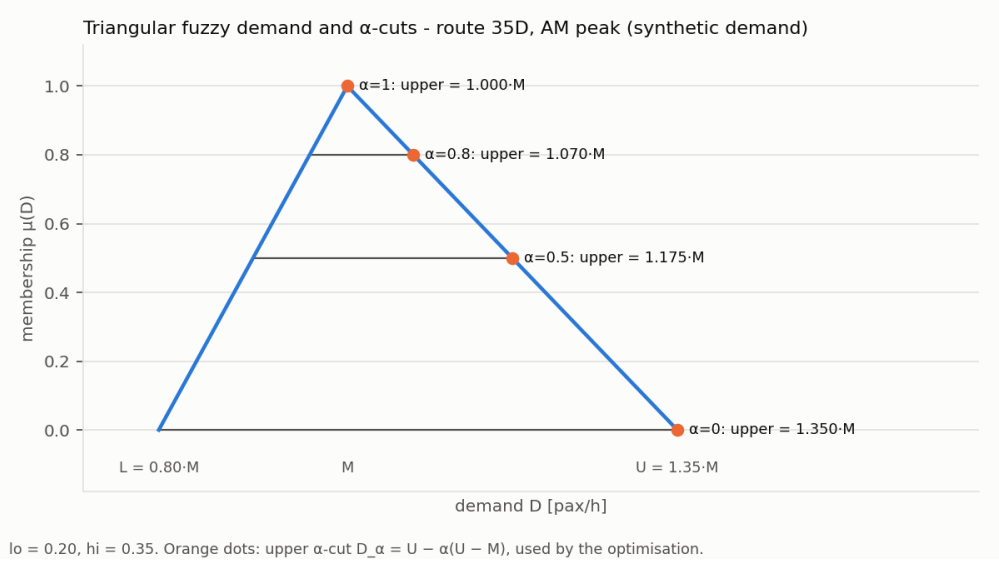


Figure 2: Triangular fuzzy demand and its α-cuts for one route-period

3.5 Goal Programming with Priorities

Each objective becomes a goal with a target taken from the published schedule: G1 waiting at most 4% below today, G2 zero overflow, G3 vehicle-km at least 3% below today, and G4 no headway above 60 minutes in the peaks or 90 minutes at midday. Deviation variables measure how far each goal is missed, and only the unwanted deviations are minimised. Preemptive goal programming minimises them in a strict priority order, locking each level before moving to the next; weighted goal programming minimises a sum of deviations divided by their baseline values; and a Chebyshev variant minimises the largest normalised deviation. The fleet and frequency-fleet constraints stay hard throughout. Three priority orders were studied (passenger-first, operator-first, equity-first), plus the balanced weighted form and the Chebyshev form. The targets for G1 and G3 were set inside the attainable range after the best achievable values were known; this is disclosed in Section 5.3.

3.6 Exact Pareto Front (AUGMECON)

The exact front was computed with AUGMECON [8]: minimise overflow Z_3 subject to waiting $Z_1 \leq \varepsilon_1$ and vehicle-km $Z_2 \leq \varepsilon_2$ on a 20×20 grid, with slack terms in the objective so that only strictly efficient points are returned, and early exit once a grid cell is infeasible. Every point of the front is an optimal solution of the same MILP, so the front and the goal-programming solutions can be compared directly. Heuristic search (NSGA-II [7]) is left as future work; the exact front computed here is the benchmark against which it would be judged (Chapter 6).

3.7 Sampling

One kind of sampling is used, and two more are planned.

- **Demand construction.** Each route receives a score proportional to its length and raised for serving PGI, ISBT-43 or ISBT-17, for terminating outside the city sectors, for terminating

at Derabassi, and for every hub stop served. The score is multiplied by a fixed log-normal noise factor (seed 42). Scores are scaled so that the network total is 81,326 boardings per day, a figure scaled from a CTU-wide ridership report that has not yet been verified. Period multipliers are 1.4 (AM), 0.8 (Midday) and 1.3 (PM).

- **Planned Monte Carlo demand days.** Monte Carlo evaluation of the schedules over sampled demand days, with demand drawn from the triangular distributions, is future work (Chapter 6).
- **Planned field sampling.** Manual boarding counts at 5 to 10 hub stops (ISBT-17, ISBT-43, PGI and others) on at least 10 weekdays, together with CTU ticket-machine data if granted, to estimate L, M and U as the 10th, 50th and 90th percentiles of observed demand.

3.8 Tools, Software and Equipment Used

Table 3: Tools, software and equipment

Category	Tool / version	Purpose
Programming	Python 3.13; pandas 2.3, NumPy 2.3, SciPy 1.16, Matplotlib 3.10, openpyxl 3.1	Data handling, analysis, figures
Optimization	PuLP 4.0.1 with the HiGHS 1.15 MILP solver	Exact MILP, goal programming, AUGMECON
Testing	pytest 9.1	Correctness of data, model and solvers
Literature	Crossref metadata API, TRB TRID, publisher pages	Verification of every reference
Hardware	Standard laptop (12 logical cores)	All runs
Equipment	None required so far	Manual boarding counts (planned) need only counting sheets

Compared with the synopsis, Google OR-Tools, PyGMO and scikit-fuzzy were not needed: PuLP with HiGHS solved the MILP directly, and the fuzzy α-cut is a closed-form expression. A metaheuristic library (DEAP, for NSGA-II) remains in the plan for the future work of Chapter 6.

Chapter 4 Work Completed

4.1 Project Timeline

The project was executed in the stages below. Each stage ended with a definition-of-done check and automated tests.

Table 4: Project phases and status

#	Phase	Main output	Status
1	Data audit and calibration	Validated route table; calibration report	Completed
2	Time periods, demand model, fuzzy numbers	Demand table (L, M, U); demand report	Completed
3	MILP formulation and baseline evaluation	Model, baseline metrics	Completed
4	Goal programming with priorities	Five scenarios $\times$ 5 α values $\times$ 2 bound settings	Completed
5	Exact Pareto front (AUGMECON)	Exact front of 46 efficient points	Completed
6	Self-review and literature verification	Examiner-style review; verified references	Completed
7	Real ridership data and validation	ETM data request; hub counts	Pending
8	Remaining model refinements	Benchmark rule, route speeds, waiting model	Planned
9	Metaheuristics and Monte Carlo simulation	NSGA-II, PSO/QPSO, simulated demand days	Future work

4.2 Work Done So Far

- **Data.** 68 routes, 389 buses, 1,662 km and 649 stops were extracted and checked against the timetable PDF; the operating windows and period activity were cross-checked (AM 64, Midday 65, PM 62 active routes).
- **Calibration.** Cycle-time scenarios were compared against the fleet; the one-way reading at 22 km/h with 10% layover was adopted, pending CTU confirmation.
- **Model.** The MILP was implemented and verified against brute-force enumeration on a small instance, with unit tests on every formula.
- **Goal programming.** Preemptive, weighted and Chebyshev goal programming was implemented; 50 scenario runs all solved to proven optimality, and the lexicographic order was verified.
- **Pareto analysis.** An exact AUGMECON front (46 efficient points) was computed from the MILP, and the goal-programming solutions were located against it.

- **Quality.** Automated tests pass for the data, model and solver code. An examiner-style self-review was carried out; its most critical point, the choice of baseline, is taken up under future work (fairer comparisons).

4.3 Challenges Faced

- **No ridership data.** Demand had to be constructed with documented assumptions; every result is therefore illustrative.
- **Infeasibility of the stated brief.** Published headways such as 35, 50 or 80 minutes were not in the headway menu; a 60-minute peak cap conflicted with published 75–90-minute headways; and 10% spares left too few buses for the published minimum frequencies. Each was diagnosed precisely and resolved with a documented change.
- **Calibration ambiguity.** Whether the published route length is one-way or a full cycle changes the fleet requirement by a factor of almost two; it was resolved by a consistency test and still needs CTU confirmation.
- **Software changes.** PuLP 4 no longer bundles the CBC solver and changed its interface; the HiGHS solver was used instead.
- **Reference verification.** Three references in the synopsis (Daduna 2004, TfL 2024, LTA 2023) could not be verified and were replaced by verified sources.

Chapter 5 Results and Discussion

All results that depend on passenger demand are illustrative. They show how the method behaves; they are not measured CTU outcomes. All numbers in this chapter are exact optimal values of the MILP, evaluated at the stated α; none comes from a simulation.

5.1 Observations

Calibration

Table 5 compares the buses needed to run the published mid-range frequencies with the 389 buses allocated. If the published length were a full cycle (S1), only 44.2% of the fleet would be needed and 57 of 68 routes would hold more buses than even their highest published frequency requires. Under the one-way reading with 10% layover, 87.6% of the fleet is needed (96.4% with 10% spares), and only 3 routes look short of buses. At 18 km/h, 23 routes could not run their published minimum frequency, which contradicts the timetable CTU actually operates. The speed that exactly fills the fleet is 19.3 km/h without spares and 21.2 km/h with 10% spares, consistent with measured Indian bus speeds [22].

Table 5: Buses needed to run the published timetable, by calibration scenario

Scenario	Speed, layover	Buses needed	Share of 389	Routes over-supplied
S1: length = full cycle	22 km/h, no layover	171.9	44.2%	57
S2: length one-way	22 km/h, no layover	309.9	79.7%	33
S2	22 km/h, 10% layover	340.9	87.6%	13
S3: S2 + 10% spares	22 km/h, 10% layover	375	96.4%	13
S2	18 km/h, no layover	378.7	97.4%	2

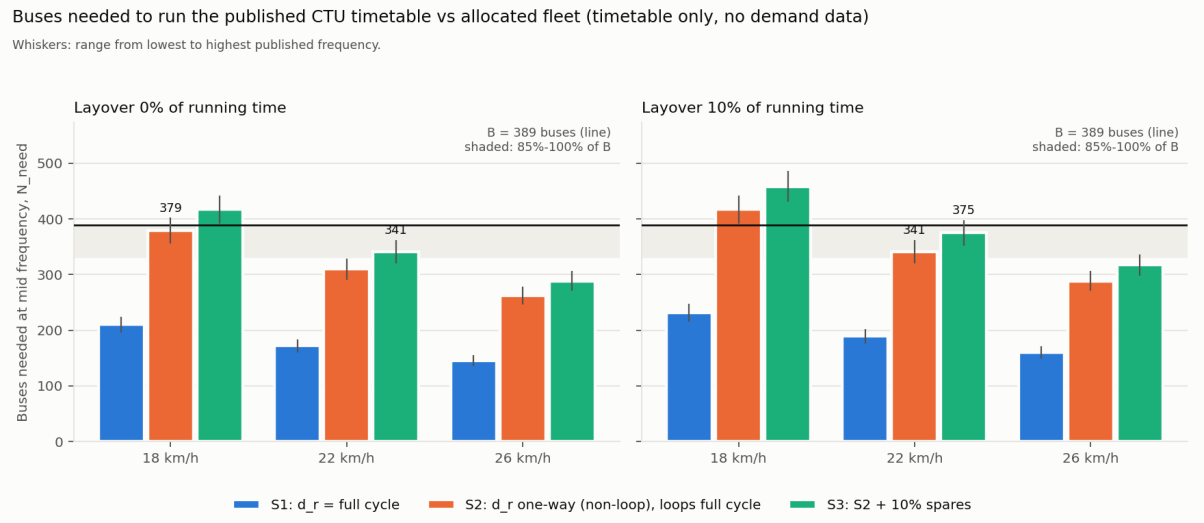


Figure 3: Buses needed across calibration scenarios vs the allocated fleet

Demand pattern

Mean modal demand is 70 passengers per hour per route in the AM peak, 39.4 at midday and 65.9 in the PM peak, with a coefficient of variation across routes of 0.39–0.41. Against published capacity, 11 route-periods are overloaded in the AM peak, mostly long routes with 75–90-minute headways. This pattern follows directly from the demand construction, in which demand grows with route length.

Baseline against the model's constraints

The published schedule keeps 385 buses on the road at the busiest time, more than the 357 allowed by the model's 8% spare rule (by 27, 28 and 23 buses in the AM, Midday and PM periods). Routes 25/25A, 40, 76 cannot run their mid-range frequency with their own buses at 22 km/h. The optimised schedules therefore have fewer buses available than the published one.

5.2 Graphs and Comparisons

Goal programming

With the published frequency limits, all five scenarios give the same schedule (waiting $+0.8\%$, overflow -0.6%): the goals do not conflict, because the overcrowded routes are already at their highest published frequency. When the maximum frequency is relaxed by two menu levels, the priorities matter (Table 6). Passenger-first and Chebyshev schedules give the largest overflow reduction but leave 100 minutes of headway above the equity caps; equity-first and balanced schedules remove every headway above the cap for a slightly smaller overflow reduction, and meet three of the four goals.

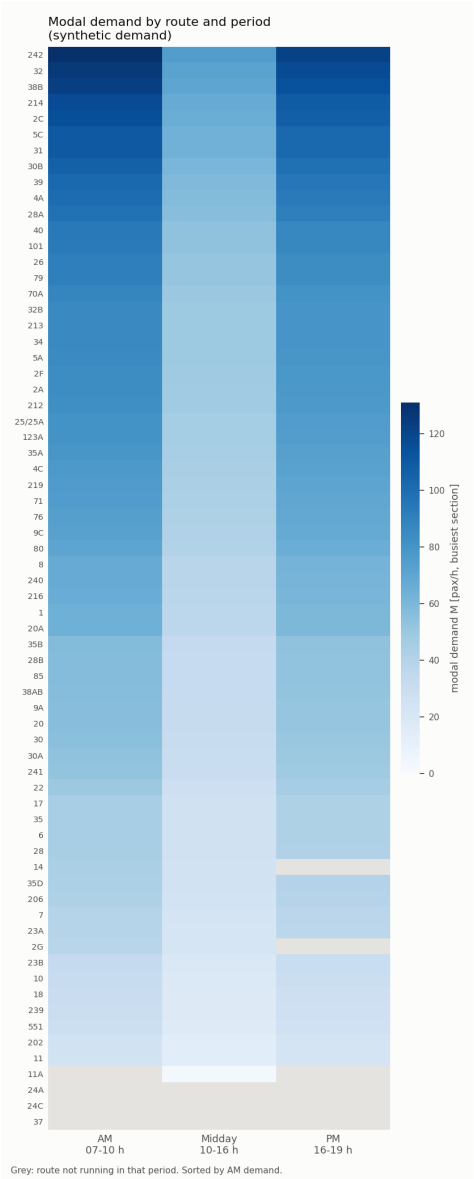


Figure 4: Modal demand by route and period

Table 6: Goal-programming scenarios ($f_{\max}$ relaxed, $\alpha = 0.5$; change vs the published schedule)

Scenario	Priority	Waiting	Vehicle-km	Overflow	Above cap (min)	Goals met
P-PASSENGER	G2 > G1 > G4 > G3	−4.0%	−2.5%	−29.6%	100	0 of 4
P-OPERATOR	G3 > G2 > G1 > G4	−3.8%	−3.0%	−29.6%	100	1 of 4
P-EQUITY	G4 > G2 > G1 > G3	−4.0%	−3.1%	−26.4%	0	3 of 4
P-BALANCED	equal normalised weights	−4.1%	−3.1%	−26.4%	0	3 of 4
CHEBYSHEV	min-max of normalised deviations	−3.8%	−3.0%	−29.6%	100	0 of 4

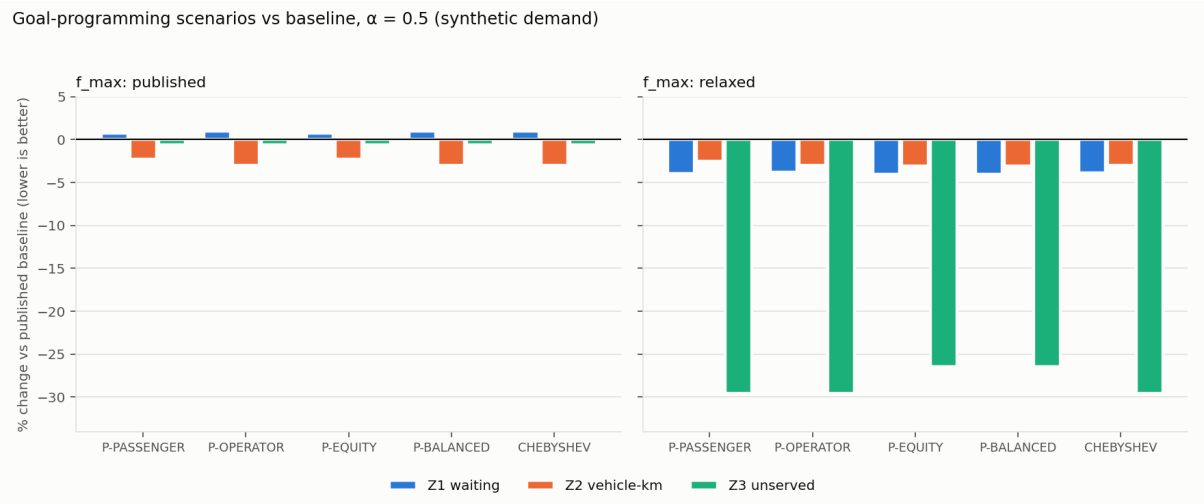


Figure 5: Change vs the published schedule by goal-programming scenario

Exact Pareto front

The 20×20 AUGMECON grid gave 298 optimal cells, 19 infeasible cells and 83 cells skipped by early exit, yielding 46 distinct strictly efficient points in 60 s. Figure 6 shows the front as two-dimensional projections, together with the goal-programming solutions and the published baseline. P-PASSENGER lies exactly on the front; P-BALANCED is 0.068 away in normalised units because it gives up a little overflow for full headway equity, a fourth criterion that the three-objective front does not show. The published baseline lies well off the front in every projection, and it breaks the fleet limit.

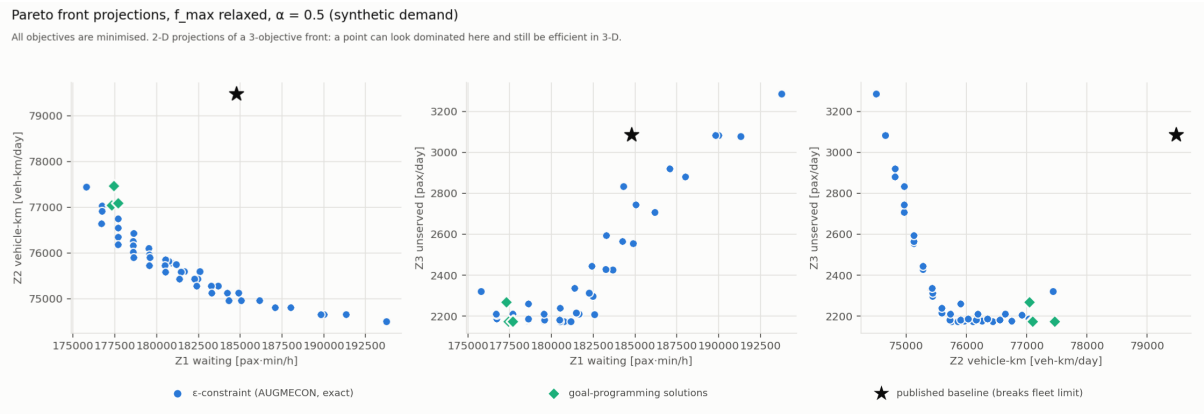


Figure 6: Two-dimensional projections of the exact (AUGMECON) front, with the goal-programming solutions and the baseline

5.3 Discussion and Limitations

The method works as intended: the model is linear and solves exactly at full network size, goal programming exposes how priorities change the schedule, and the exact Pareto front shows the trade-offs among waiting, vehicle-km and overflow. Under synthetic demand, the optimised schedules reduce waiting and overflow relative to the published schedule while using fewer buses. Several caveats limit what the numbers mean.

- **Synthetic demand.** Demand grows with route length and with hub, PGI and ISBT flags, so the gains concentrate on long, infrequent routes partly by construction.
- **Calibration.** A single speed of 22 km/h makes some routes look over-supplied, which frees buses for others; route-specific running times are needed to test this.
- **Targets.** The goal-programming targets were set after the best achievable values were known, so goal achievement is conditional on those targets.
- **No stress test yet.** The results are deterministic values of the model at $\alpha = 0.5$. They have not been stress-tested against day-to-day demand variation or irregular arrivals, which is planned (Chapter 6).
- **Modelling simplifications.** The waiting objective is a per-hour rate not weighted by period length; the $h/2$ waiting rule overstates waiting at long headways; there is no within-period peak-hour factor; buses move freely between routes across periods; and there are no transfers, depot, crew or stop-capacity constraints.
- **Relaxed frequency limits.** The headline results allow two headway levels shorter than the published minimum, which would need CTU approval.

Chapter 6 Future Work with Timeline

The remaining work focuses on replacing assumptions with data, on extending the exact linear methods with heuristic search and simulation, and on testing whether the gains survive fairer comparisons.

- **Real ridership.** CTU ticket-machine (ETM) records and manual boarding counts at hub stops; estimate L, M and U as the 10th, 50th and 90th percentiles of observed daily demand.
- **Metaheuristic search.** NSGA-II (a genetic algorithm) [7] will approximate the Pareto front in a single run and will be benchmarked against the exact AUGMECON front by hypervolume, generational distance, inverted generational distance, coverage and a strict-dominance check. Particle Swarm Optimization (PSO) and Quantum-behaved PSO (QPSO) will then be evaluated against NSGA-II and the MILP baseline for solution quality, convergence, computational time and robustness, as alternatives for large-scale problems and for routing and scheduling extensions.
- **Monte Carlo simulation and sensitivity.** Schedules will be evaluated over sampled demand days (demand drawn from the triangular distributions with a common day-level shock, bus arrivals with gamma-distributed headways, expected wait $(h/2)(1+\mathrm{CV}^2)$), with one-factor sensitivity to speed, capacity, spares, layover, α and fleet size, and an out-of-sample test.
- **Fairer comparisons.** A simple benchmark rule, route-specific running times, and a sweep of goal targets.
- **Better waiting model.** Passengers time their arrivals at long headways, and demand peaks within each period.
- **Operations.** Depot and interlining constraints, driver duties, stop and terminal capacity at shorter headways, and timetable-level scheduling of departure times.
- **Demand prediction.** Once labelled ridership exists, a prediction model (for example gradient-boosted trees such as XGBoost) could forecast L, M and U for the same α-cut model.

Table 7: Timeline for the remaining work

Period	Planned work
Month 1	Request ETM ticket data from CTU; boarding counts at 5–10 hubs; confirm the route-length convention and fleet composition
Month 2	Simple benchmark (demand-proportional and square-root frequency rules); route-specific running times; arrival-timing waiting model; peak-hour factor; NSGA-II benchmark against the exact front
Month 3	Replace synthetic demand with observed L/M/U; re-run all stages; Monte Carlo simulation and sensitivity analysis; robustness to demand construction and goal targets
Month 4	PSO/QPSO exploration if time permits; final report and viva; conference paper preparation and submission

6.1 Conference Targeted

- **Conference of Transportation Research Group of India (CTRG).** Biennial; the 8th edition was held at IIT Guwahati in December 2025, so the next edition is expected in 2027. The call for papers must be confirmed on the official site.
- **Annual Convention of the Operational Research Society of India (ORSI).** Usually held in December; the date and deadlines of the next convention must be confirmed on the society's site.

A submission would present the method and its results as a methodological case study, and would be strengthened substantially by the real-ridership validation planned in Months 1–3.

References

[1] A. Ceder, *Public Transit Planning and Operation: Modeling, Practice and Behavior*, 2nd ed. Boca Raton, FL: CRC Press, 2016, doi: 10.1201/b18689.

[2] A. Ceder and N. H. M. Wilson, "Bus network design," *Transportation Research Part B*, vol. 20, no. 4, pp. 331–344, 1986, doi: 10.1016/0191-2615(86)90047-0.

[3] P. G. Furth and N. H. M. Wilson, "Setting frequencies on bus routes: theory and practice," *Transportation Research Record*, no. 818, pp. 1–7, 1981.

[4] O. J. Ibarra-Rojas, F. Delgado, R. Giesen and J. C. Muñoz, "Planning, operation, and control of bus transport systems: A literature review," *Transportation Research Part B*, vol. 77, pp. 38–75, 2015, doi: 10.1016/j.trb.2015.03.002.

[5] H. Martínez, A. Mauttone and M. E. Urquhart, "Frequency optimization in public transportation systems: Formulation and metaheuristic approach," *European Journal of Operational Research*, vol. 236, no. 1, pp. 27–36, 2014, doi: 10.1016/j.ejor.2013.11.007.

[6] İ. Ö. Verbas and H. S. Mahmassani, "Exploring trade-offs in frequency allocation in a transit network using bus route patterns: Methodology and application to large-scale urban systems," *Transportation Research Part B*, vol. 81, pp. 577–595, 2015, doi: 10.1016/j.trb.2015.06.018.

[7] K. Deb, A. Pratap, S. Agarwal and T. Meyarivan, "A fast and elitist multiobjective genetic algorithm: NSGA-II," *IEEE Transactions on Evolutionary Computation*, vol. 6, no. 2, pp. 182–197, 2002, doi: 10.1109/4235.996017.

[8] G. Mavrotas, "Effective implementation of the ε-constraint method in multi-objective mathematical programming problems," *Applied Mathematics and Computation*, vol. 213, no. 2, pp. 455–465, 2009, doi: 10.1016/j.amc.2009.03.037.

[9] G. Mavrotas and K. Florios, "An improved version of the augmented ε-constraint method (AUGMECON2) for finding the exact Pareto set in multi-objective integer programming problems," *Applied Mathematics and Computation*, vol. 219, no. 18, pp. 9652–9669, 2013, doi: 10.1016/j.amc.2013.03.002.

[10] A. Charnes and W. W. Cooper, *Management Models and Industrial Applications of Linear Programming*, vols. 1–2. New York: Wiley, 1961.

[11] D. Jones and M. Tamiz, *Practical Goal Programming*. New York: Springer, 2010, doi: 10.1007/978-1-4419-5771-9.

[12] L. A. Zadeh, "Fuzzy sets," *Information and Control*, vol. 8, no. 3, pp. 338–353, 1965, doi: 10.1016/S0019-9958(65)90241-X.

[13] H.-J. Zimmermann, "Fuzzy programming and linear programming with several objective functions," *Fuzzy Sets and Systems*, vol. 1, no. 1, pp. 45–55, 1978, doi: 10.1016/0165-0114(78)90031-3.

[14] M. Inuiguchi and J. Ramík, "Possibilistic linear programming: a brief review of fuzzy mathematical programming and a comparison with stochastic programming in portfolio selection problem," *Fuzzy Sets and Systems*, vol. 111, no. 1, pp. 3–28, 2000, doi: 10.1016/S0165-0114(98)00449-7.

[15] K. An and H. K. Lo, “Two-phase stochastic program for transit network design under demand uncertainty,” *Transportation Research Part B*, vol. 84, pp. 157–181, 2016, doi: 10.1016/j.trb.2015.12.009.

[16] K. Gkiotsalitis, M. Schmidt and E. van der Hurk, “Subline frequency setting for autonomous minibusses under demand uncertainty,” *Transportation Research Part C*, vol. 135, art. 103492, 2022, doi: 10.1016/j.trc.2021.103492.

[17] R. O. Arbex and C. B. da Cunha, “Efficient transit network design and frequencies setting multi-objective optimization by alternating objective genetic algorithm,” *Transportation Research Part B*, vol. 81, pp. 355–376, 2015, doi: 10.1016/j.trb.2015.06.014.

[18] S. B. Jha, J. K. Jha and M. K. Tiwari, “A multi-objective meta-heuristic approach for transit network design and frequency setting problem in a bus transit system,” *Computers & Industrial Engineering*, vol. 130, pp. 166–186, 2019, doi: 10.1016/j.cie.2019.02.025.

[19] Z. Ahern, A. Paz and P. Corry, “Approximate multi-objective optimization for integrated bus route design and service frequency setting,” *Transportation Research Part B*, vol. 155, pp. 1–25, 2022, doi: 10.1016/j.trb.2021.10.007.

[20] W. Leong, K. Goh, S. Hess and P. Murphy, “Improving bus service reliability: The Singapore experience,” *Research in Transportation Economics*, vol. 59, pp. 40–49, 2016, doi: 10.1016/j.retrec.2016.07.025.

[21] Transport for London, *Guidelines for Planning Bus Services*. London: TfL, Aug. 2012. [Online]. Available: https://foi.tfl.gov.uk/FOI-2321-1617/bus-service-planning-guidelines.pdf

[22] S. Mahesh and G. Ramadurai, “Analysis of driving characteristics and estimation of pollutant emissions from intra-city buses,” *Transportation Research Procedia*, vol. 27, pp. 1211–1218, 2017, doi: 10.1016/j.trpro.2017.12.071.

[23] G. V. Jain, S. S. Jain and M. Parida, “Evaluation of travel speed of conventional buses and bus rapid transit service in Ahmedabad city, India using geo-informatics,” *Journal of Public Transportation*, vol. 24, art. 100034, 2022, doi: 10.1016/j.jpubtr.2022.100034.

[24] R. Fedujwar and A. Agarwal, “A systematic review on crowding valuation in public transport,” *Public Transport*, vol. 16, no. 3, pp. 743–773, 2024, doi: 10.1007/s12469-024-00363-w.

[25] Chandigarh Transport Undertaking, “Time Table of CTU Bus Services: Local and Tri-City Routes, Depot Nos. II, III and IV,” as on 25.07.2026. [Online]. Available: chdctu.gov.in